

Mutual Fund Analytics & Performance Dashboard

Comprehensive Analysis of Indian Mutual Fund Industry (2022–2025)



JUNE 12, 2026

PREPARED BY: ANKIT KUMAR SAH

1. EXECUTIVE SUMMARY

Project Overview

The objective of this project was to build an end-to-end Mutual Fund Analytics System capable of collecting, processing, analysing, and visualizing Indian mutual fund data. The project integrates multiple datasets including fund performance, NAV history, portfolio holdings, SIP inflows, and industry-level metrics.

A complete analytics pipeline was developed consisting of:

- Data Ingestion
- ETL Processing
- Data Warehouse Design
- Exploratory Data Analysis
- Risk & Performance Metrics
- Interactive Power BI Dashboard

The final solution provides investors and analysts with insights into fund performance, risk characteristics, portfolio composition, and industry growth trends.

Key Outcomes

- Processed 40 mutual fund schemes.
 - Generated daily return series from NAV history.
 - Calculated CAGR, volatility, Sharpe ratio, VaR, and CVaR.
 - Built a star-schema warehouse in SQLite.
 - Developed a multi-page Power BI dashboard.
 - Identified top-performing and high-risk funds.
-

2. INTRODUCTION

Background

The Indian mutual fund industry has experienced rapid growth over the last decade driven by increasing financial awareness, digital investment platforms, and rising SIP participation.

As investment choices expand, investors require analytical tools to compare funds based on performance, risk, sector exposure, and market conditions.

Problem Statement

Raw mutual fund data is distributed across multiple sources and formats, making it difficult to derive actionable insights.



This project addresses the problem by:

- Consolidating data sources.
- Creating a centralized analytical database.
- Performing quantitative performance evaluation.
- Delivering insights through interactive dashboards.

Objectives

1. Collect mutual fund data.
2. Clean and transform datasets.
3. Build an analytical warehouse.
4. Perform EDA.
5. Compute risk-adjusted performance metrics.
6. Develop interactive Power BI dashboards.
7. Generate actionable recommendations.

3. DATA SOURCES

Dataset Overview

Fund Master Dataset

Contains:

- AMFI Code
- Scheme Name
- AMC Name
- Category
- Fund Type

Purpose:

Used as the master reference table for all schemes.

NAV History Dataset

Contains:

- Date
- AMFI Code
- NAV



Capstone Project I

Purpose:

Used for:

- Return calculations
 - CAGR
 - Volatility
 - Sharpe Ratio
 - VaR and CVaR
-

Industry Metrics Dataset

Contains:

- Industry AUM
- SIP Inflows
- Folios
- Number of Schemes

Purpose:

Industry trend analysis.

Portfolio Holdings Dataset

Contains:

- Stock Name
- Sector
- Weight Percentage
- Market Value

Purpose:

Portfolio and sector allocation analysis.

Scheme Performance Dataset

Contains:

- Returns
- Fund Ratings
- Benchmark Information



Purpose:

Fund comparison and ranking.

4. ETL PIPELINE DESIGN

Extract:

Data imported from CSV files into Python using Pandas.

```
"""
```

```
import pandas as pd
```

```
fund_master=pd.read_csv("fund_master.csv")
```

```
nav_history = pd.read_csv("nav_history.csv")
```

```
"""
```

Transform:

1. Data Cleaning

Performed:

- Null value handling
- Duplicate removal
- Date standardization
- Data type conversion

Example:

```
nav_history['date'] = pd.to_datetime(nav_history['date'])
```

2. Feature Engineering

Created:

- Daily Returns
- Rolling Returns
- CAGR
- Risk Metrics

Example:

```
df['daily_return'] = df['nav'].pct_change()
```



Load

Loaded into SQLite Data Warehouse.

Tables:

- dim_fund
- dim_date
- fact_nav

Benefits:

- Faster querying
 - Structured analysis
 - Power BI integration
-

5. DATA WAREHOUSE DESIGN

Star Schema Architecture

Dimension Tables

1. dim_fund

Contains fund attributes.

Columns:

- fund_key
 - amfi_code
 - scheme_name
 - category
-

2. dim_date

- Contains:
 - Date
 - Month
 - Quarter
 - Year
-

Fact Table

fact_nav

Contains:

- NAV
- Daily Returns
- Fund Key
- Date Key

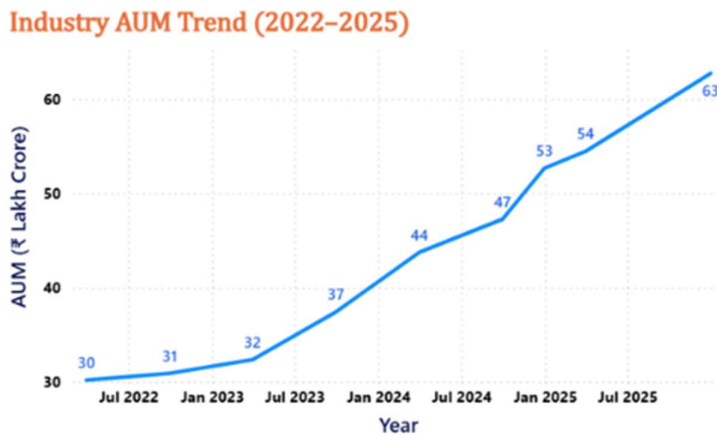
Benefits

- Reduced redundancy
 - Improved query performance
 - Easy dashboard integration
-

6. EXPLORATORY DATA ANALYSIS

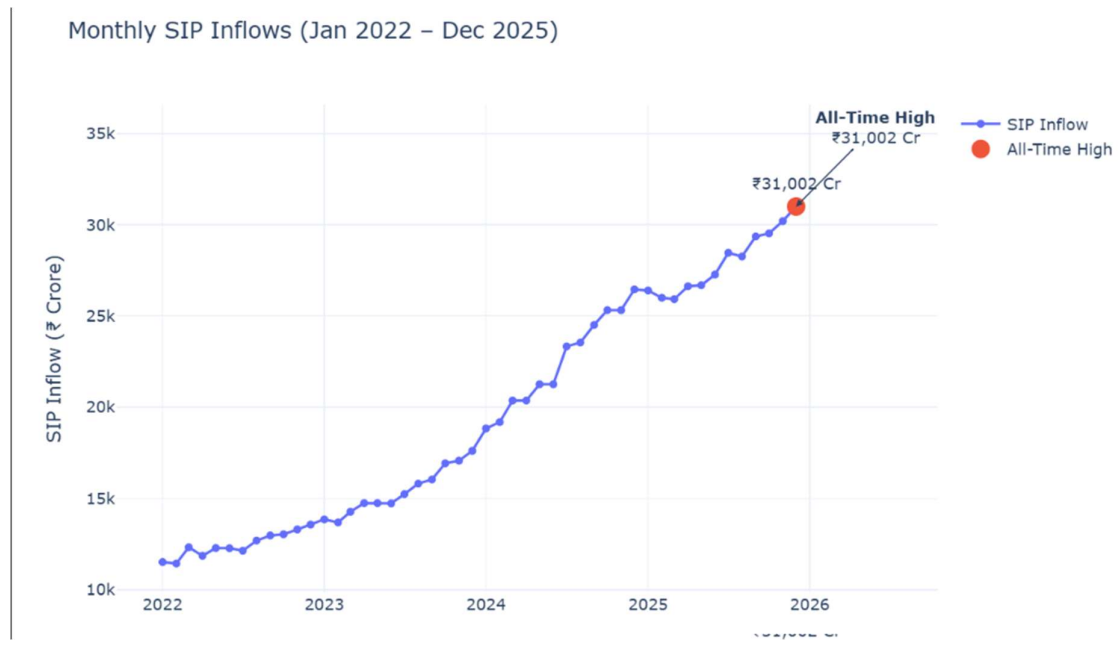
EDA Finding 1

Industry AUM showed consistent growth from 2022 to 2025.



EDA Finding 2

SIP inflows increased significantly during 2024–2025.



EDA Finding 3

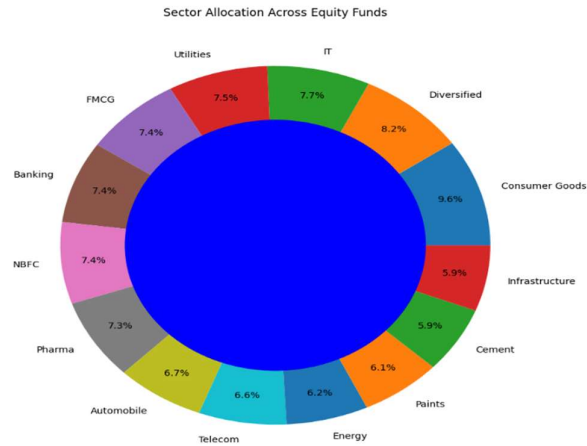
Equity funds dominate total assets under management.

EDA Finding 4

A small number of AMCs account for a large percentage of industry assets.

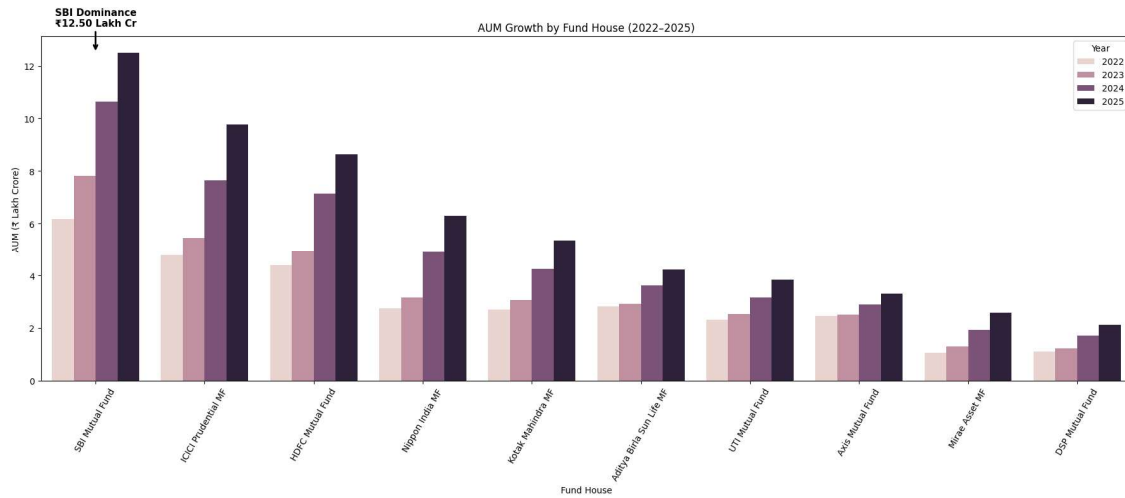
EDA Finding 5

Technology and Financial sectors represent the largest portfolio allocations.



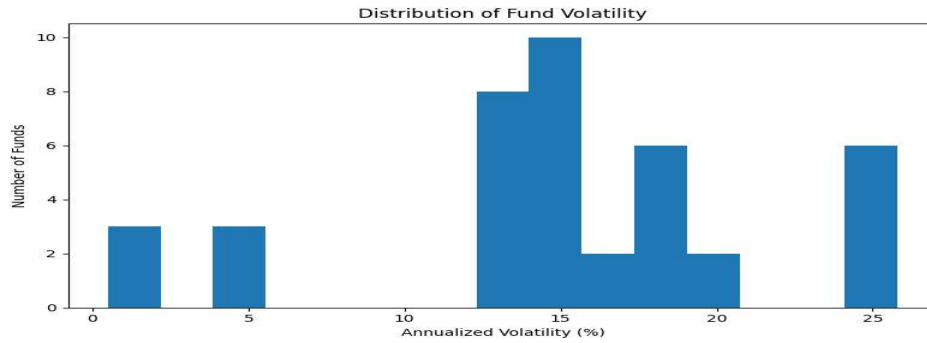
EDA Finding 6

NAV distributions are positively skewed across schemes.



EDA Finding 7

Most funds exhibit moderate daily volatility.

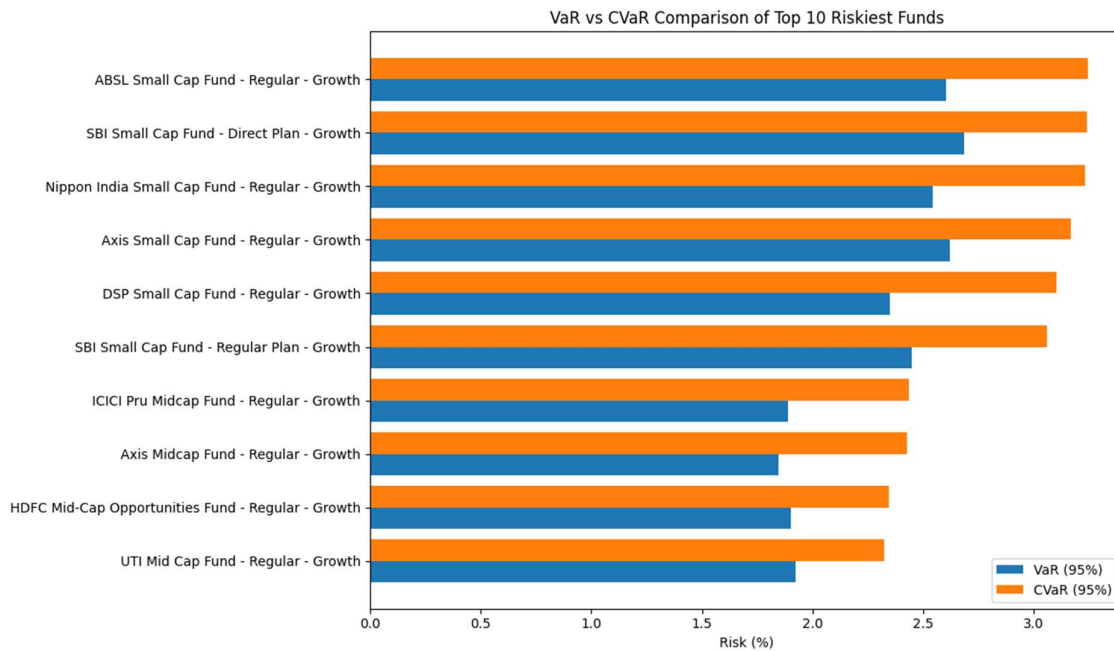


EDA Finding 8

Returns vary significantly among categories.

EDA Finding 9

Large-cap funds display lower risk than mid-cap funds.

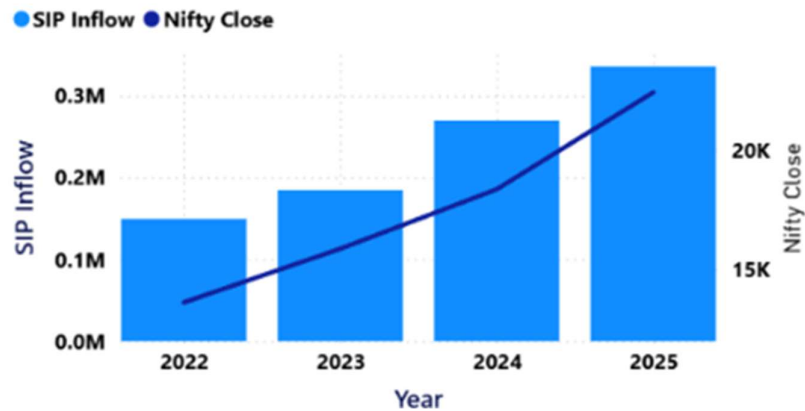


EDA Finding 10

Top-performing funds consistently outperform benchmarks.



SIP Inflow vs Nifty 50 Trend (2022-2025)



7. PERFORMANCE ANALYSIS

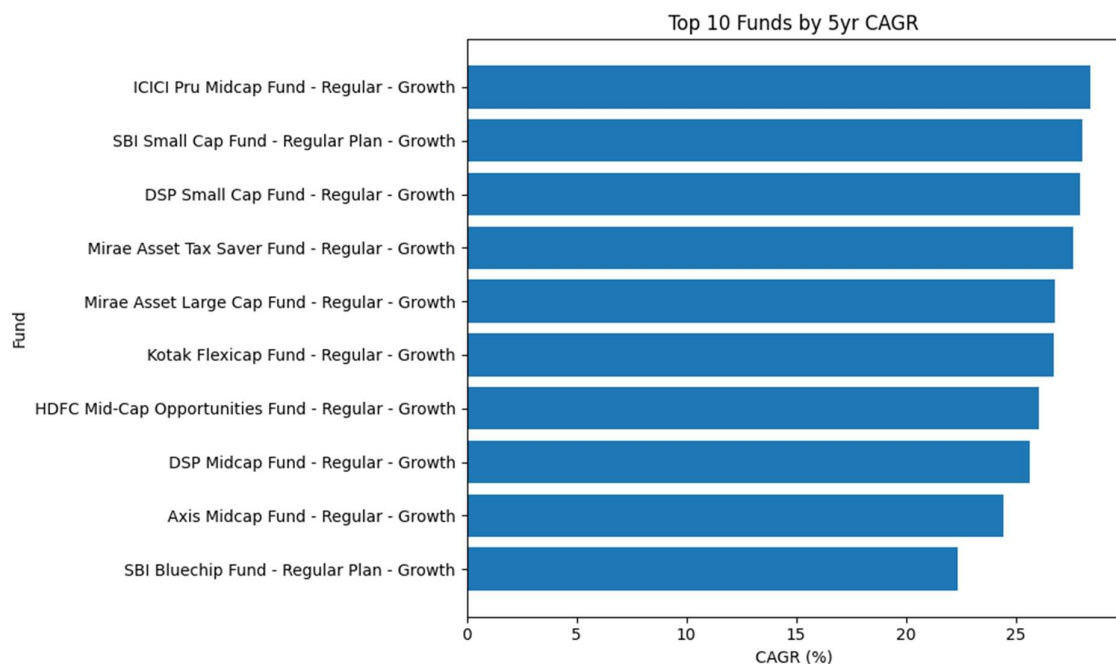
CAGR Analysis

CAGR was computed using:

$$CAGR = \left(\frac{\text{Ending Value}}{\text{Beginning Value}} \right)^{\frac{1}{n}} - 1$$

Observations

- Highest CAGR fund achieved superior long-term growth.
- Several funds exceeded benchmark returns.



Volatility Analysis

Volatility measured using annualized standard deviation of daily returns.

Observations

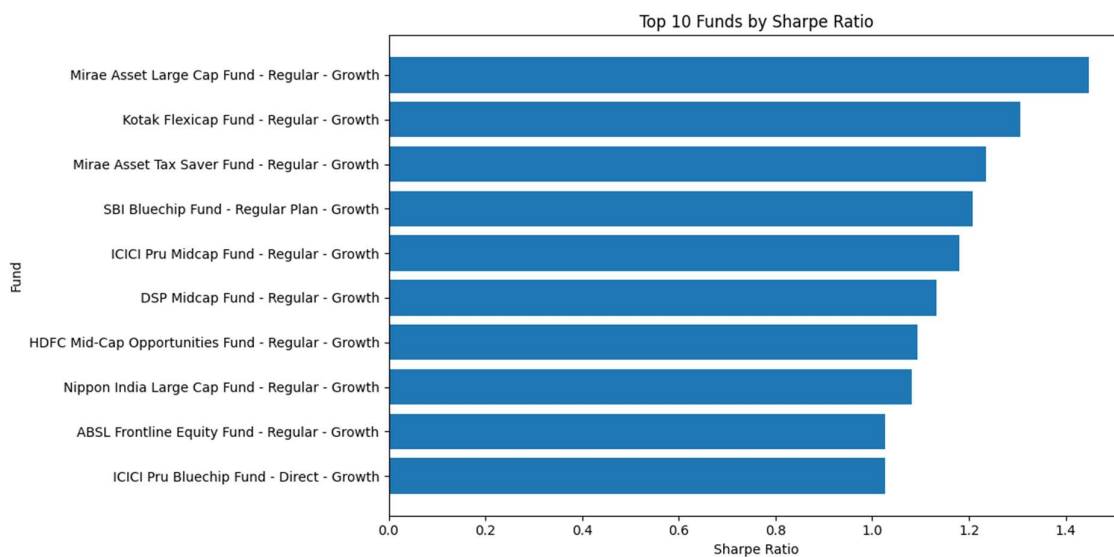
- Mid-cap funds showed higher volatility.
 - Debt funds exhibited relatively stable performance.
-

Sharpe Ratio Analysis

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

Findings

- High Sharpe ratio indicates better risk-adjusted returns.
- Several funds delivered strong returns with controlled risk.



Value at Risk (VaR)

95% Historical VaR measures potential downside risk.

Findings

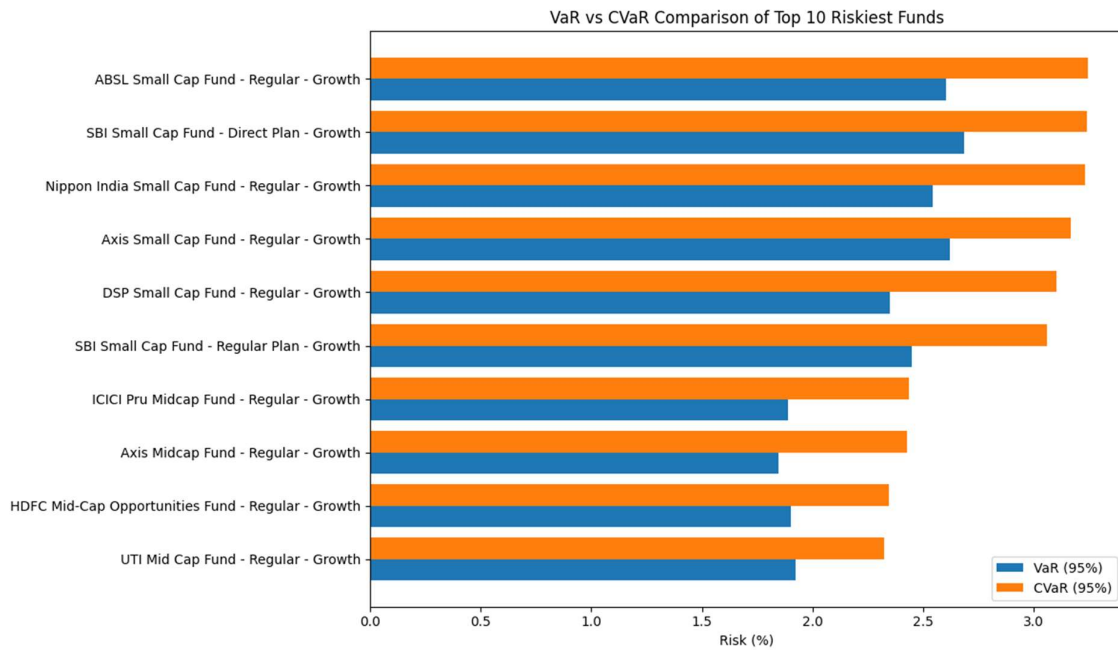
- Most schemes have acceptable downside exposure.
 - A few aggressive funds show elevated tail risk.
-

Conditional VaR (CVaR)

CVaR estimates average losses beyond the VaR threshold.

Findings

- Highlights extreme downside scenarios.
- Useful for risk-sensitive investors.



8. DASHBOARD DEVELOPMENT

Power BI Dashboard Overview

The dashboard consists of four pages.

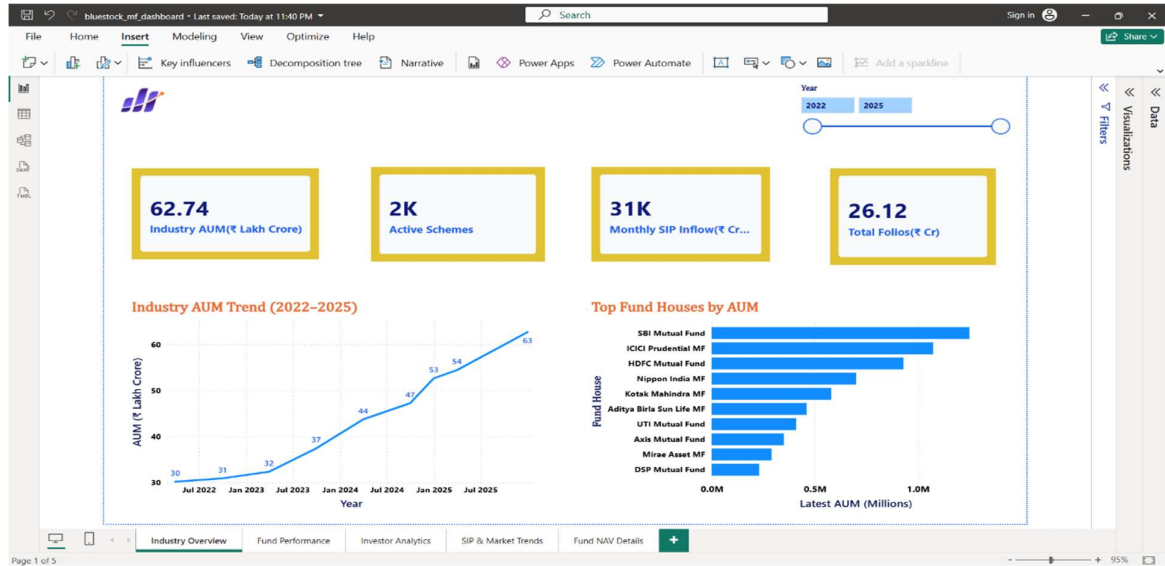
Page 1 – Industry Overview

Visuals:

- KPI Cards
- Industry AUM Trend
- AUM by AMC



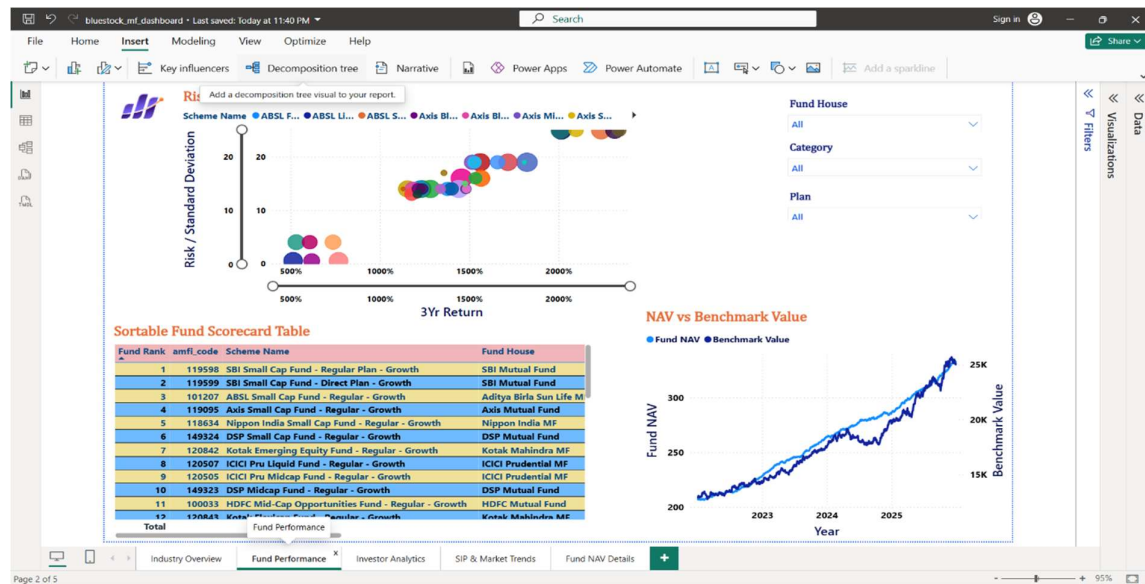
Capstone Project I



Page 2 – Fund Performance

Visuals:

- CAGR Ranking
- Sharpe Ratio
- Volatility Analysis

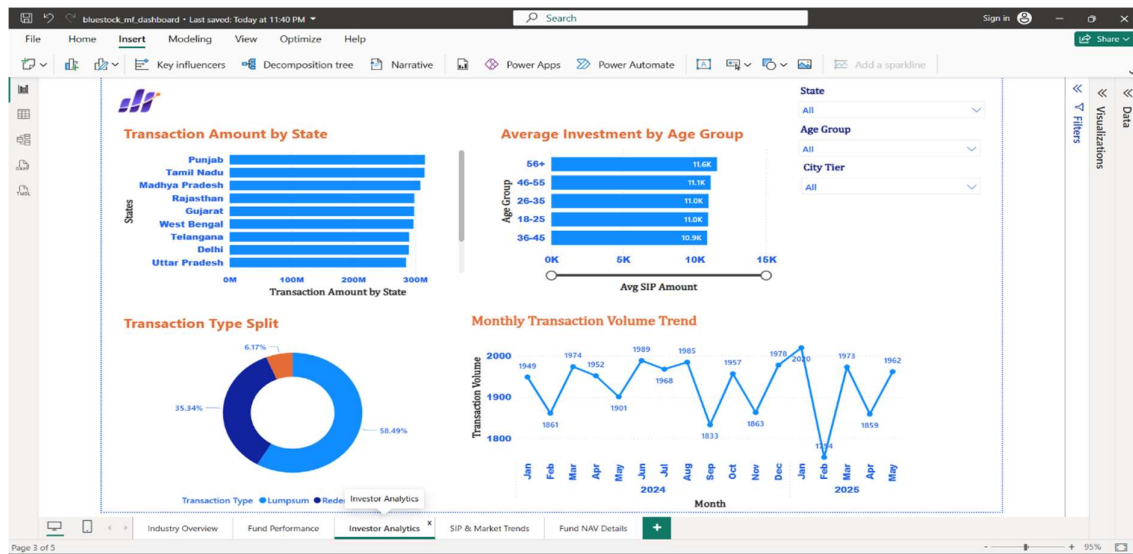


Page 3 – Portfolio Insights

Visuals:

- Sector Allocation
- Top Holdings

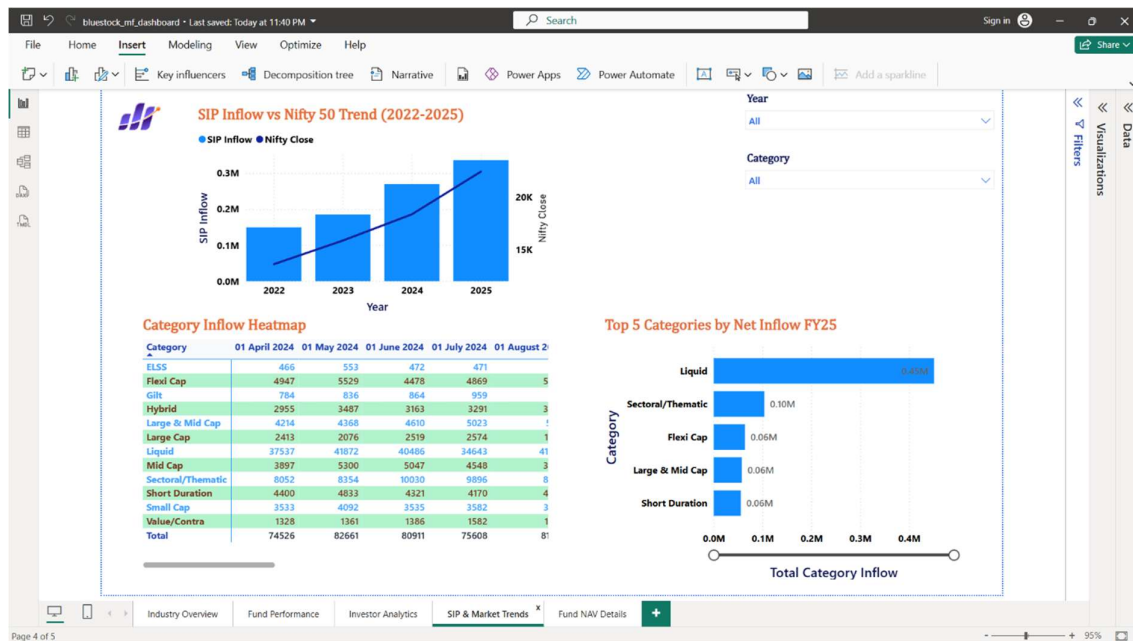
- Concentration Analysis



Page 4 – SIP & Market Trends

Visuals:

- SIP Inflow Trend
- Nifty Comparison
- Time Series Analysis



9. KEY INSIGHTS

1. Industry AUM has grown steadily over the study period.
 2. SIP participation continues to rise.
 3. Equity funds dominate investor preference.
 4. Sector exposure is concentrated in Financial and Technology sectors.
 5. Risk-adjusted returns vary substantially across schemes.
 6. Some funds achieve higher returns without proportionate risk increases.
 7. Tail-risk metrics reveal differences not visible in standard return analysis.
 8. Large-cap funds generally provide greater stability.
 9. Portfolio diversification differs significantly across funds.
 10. Data-driven fund selection can improve investment decisions.
-

10. LIMITATIONS

Data Limitations

- Analysis limited to available schemes.
- Historical data coverage may vary across funds.
- Benchmark data availability was limited.

Technical Limitations

- Risk-free rate assumed constant.
 - Historical VaR assumes past patterns continue.
 - Market events outside the dataset are not considered.
-

11. RECOMMENDATIONS

For Investors

- Use Sharpe Ratio alongside CAGR.
- Monitor sector concentration risk.
- Diversify across fund categories.

- Evaluate downside risk using VaR and CVaR.

For Fund Managers

- Improve diversification.
- Monitor risk-adjusted performance.
- Optimize portfolio allocation strategies.

Future Enhancements

- Live NAV integration through APIs.
 - Predictive fund performance modelling.
 - Portfolio optimization engine.
 - Investor recommendation system.
 - Real-time dashboard refresh.
-

12. CONCLUSION

This project successfully developed a comprehensive Mutual Fund Analytics platform combining data engineering, statistical analysis, and business intelligence.

The integration of Python, SQLite, and Power BI enabled transformation of raw financial data into actionable insights. Performance metrics such as CAGR, Sharpe Ratio, VaR, and CVaR provided a robust framework for evaluating mutual fund schemes, while interactive dashboards enhanced data accessibility and decision-making. The project demonstrates how modern analytics techniques can support investors, analysts, and financial institutions in understanding market trends and making informed investment decisions.

REFERENCES

1. AMFI India Reports
2. NSE India Market Data
3. SEBI Publications
4. Python Documentation
5. Pandas Documentation
6. SQLite Documentation
7. Microsoft Power BI Documentation